

When Does Risk-Managed Momentum Add Value?

A Reproducible, Out-of-Sample, Regime-Conditional Study of a Volatility-Targeted Dual-Momentum Strategy

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The Question

A defensive strategy is *expected* to trail equities in a bull market, so “did it beat the market?” is the wrong question. The real one is conditional:

Under what conditions does risk-managed allocation add value, and can we tell that value from luck?

Four falsifiable hypotheses:

H1 Lower volatility and drawdown than buy-and-hold.

H2 Strongest in equity-stress regimes, weakest in calm bull markets.

H3 Stable across parameters and out-of-sample.

H4 Differences are statistically distinguishable from zero.

Method

A rules-based monthly strategy over four ETFs (US equities, intl equities, bonds, cash):

1. **Absolute momentum** decides risk-on vs risk-off.

2. **Relative momentum** picks the strongest risk asset.

3. **Volatility targeting** sizes the position to a $\sigma^* = 10\%$ target:

$$w_t = \text{clip}\left(\frac{\sigma^*}{\sigma_{ji,t}}, 0, 1\right)$$

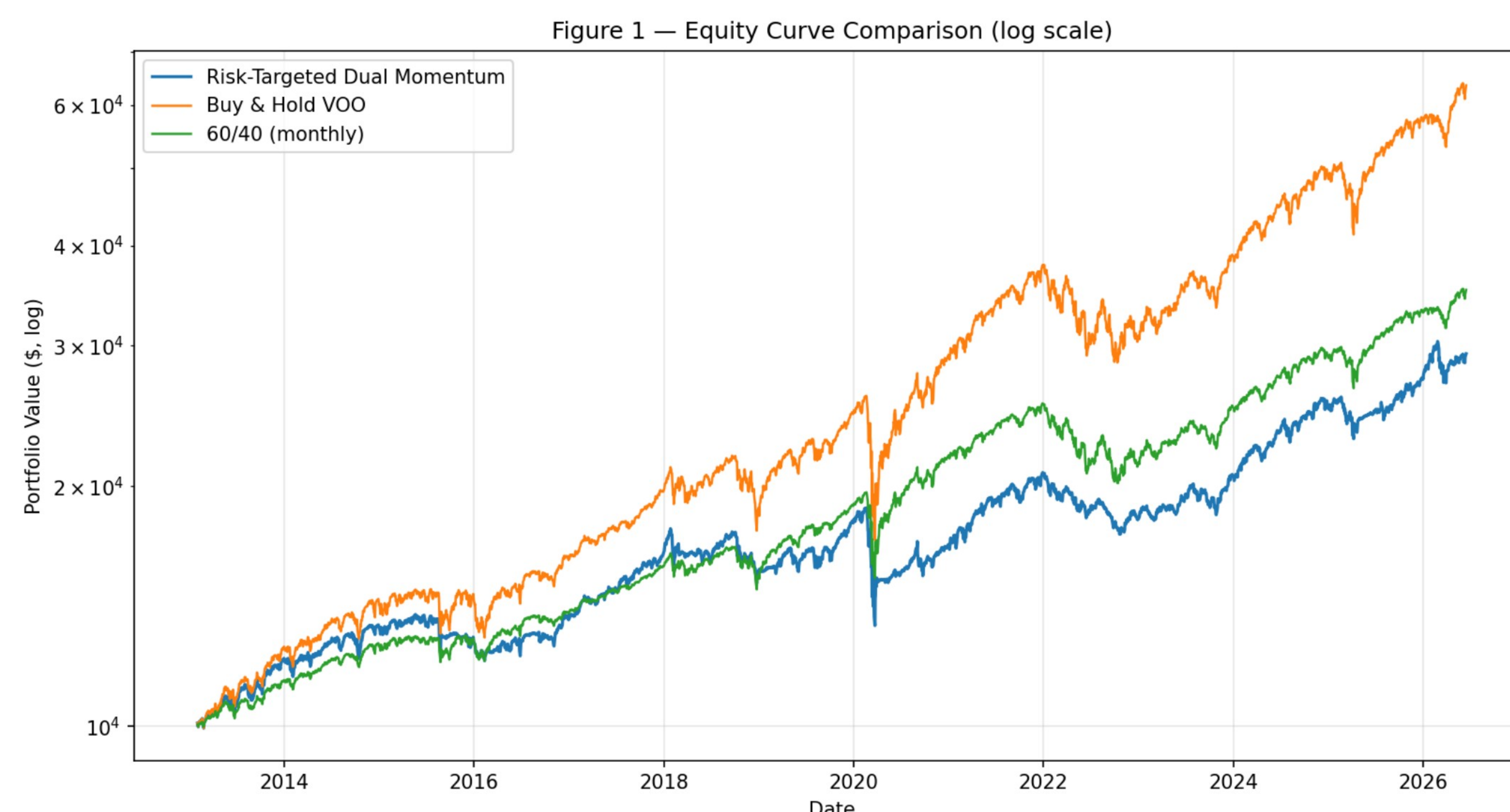
Corrected backtest: no lookahead bias, turnover-based costs, and a *real* time-varying risk-free rate. Validated five ways: in/out-of-sample split, walk-forward parameter selection, regime-conditional windows, a 36-cell sensitivity grid, and a block-bootstrap significance test. All results reproducible from cached data and a fixed seed.

Robustness & Significance

No configuration in the 36-cell grid beats a 60/40 on Sharpe: the underperformance is *structural*, not bad tuning. Bootstrap testing finds the strategy statistically *indistinguishable* from 60/40.

Walk-forward re-optimization made things worse (Sharpe 0.33 vs 0.58): a clean demonstration of the backtest-overfitting trap.

Result 1: Honest Benchmarking (2013–2026)



Portfolio	CAGR	Vol	Sharpe	MaxDD
Strategy	8.30%	12.4%	0.58	-28.8%
Buy & Hold VOO	14.75%	17.0%	0.81	-34.0%
60/40 (monthly)	9.84%	10.6%	0.79	-22.0%

The strategy lowers volatility and drawdown vs pure equity (**H1 supported**), but is dominated by a simple 60/40 on every metric. Read alone, the average calls it an inefficient risk-reducer. The conditional analysis shows that average hides the real story.

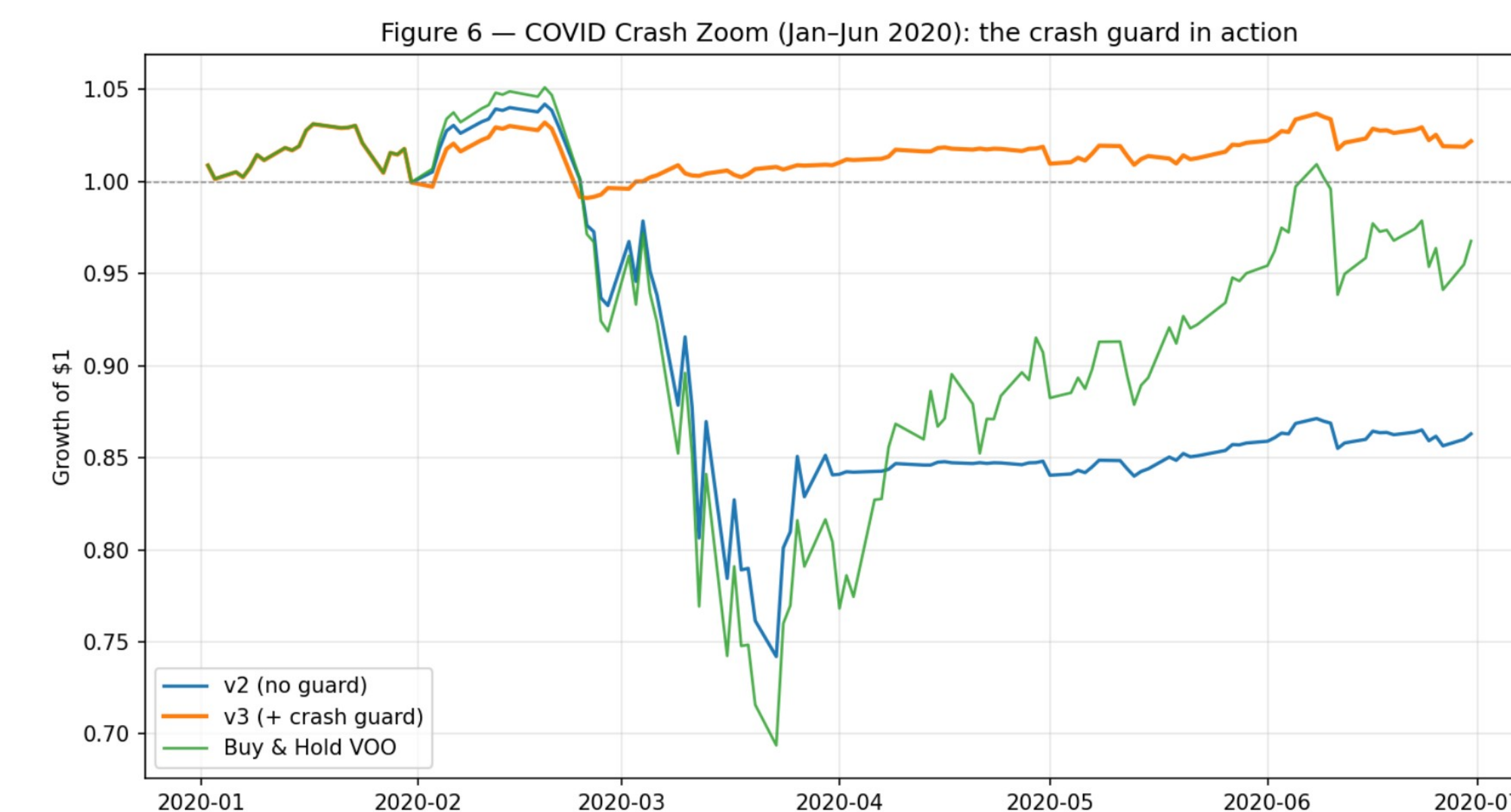
Result 2: The Value Is Regime-Conditional

Crisis window	Strategy	VOO	60/40
2018 Q4 selloff	-13.3%	-19.0%	-11.4%
2022 bear	-14.2%	-24.1%	-20.2%
COVID crash	-28.5%	-33.7%	-21.7%

In the *slow* 2022 bear it beat both benchmarks decisively. In the *fast* COVID crash it suffered its worst loss of the sample, because a monthly signal cannot react in days. **The mechanism that protects against slow bears is the same one that fails in fast crashes** (H2 supported, once conditioned).

The Fix: A Daily Crash Guard (v3)

Motivated directly by the diagnosed failure: a daily trend gauge moves the equity sleeve to cash when the fast trend breaks, applied with a one-day delay.



Window	v2 MaxDD	v3 MaxDD
2018 Q4	-13.7%	-3.0%
COVID	-28.8%	-4.0%
2022 bear	-14.6%	-12.7%

The guard sharply cuts fast-crash losses (COVID return -28.5% → -1.9%; drawdown -28.8% → -4.0%) and delivers statistically significant downside protection vs both benchmarks in down markets.

The Deeper Finding: A Tradeoff

Guard speed	Sharpe	CAGR	MaxDD
10-day	-0.24	-0.4%	-27.0%
20-day	0.17	2.6%	-22.2%
40-day	0.34	4.2%	-21.9%
60-day	0.58	6.7%	-16.0%

Faster guards protect best in crashes but whipsaw most in calm markets. **Crash protection and whipsaw are two ends of one speed dial** – the deepest result, reached only by diagnosing a weakness and engineering against it.

The contribution is not a profitable trading signal; it is a reproducible demonstration of how a financial strategy can be specified, stress-tested, falsified, repaired, and interpreted honestly.

Educational research only; not investment advice.